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riscv64-elf-objdump -t kernel/kernel | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > kernel/kernel.sym
== Test sleep, no arguments ==
$ make qemu-gdb
sleep, no arguments: OK (3.9s)
== Test sleep, returns ==
$ make qemu-gdb
sleep, returns: OK (0.6s)
== Test sleep, makes syscall ==
$ make qemu-gdb
sleep, makes syscall: FAIL (1.9s)
...
    hart 1 starting
    init: starting sh
GOOD $ sleep 10
    $ echo FAIL
BAD  FAIL
    $ qemu-system-riscv64: terminating on signal 15 from pid 75020 (<unknown process>)
unexpected lines in output
QEMU output saved to xv6.out.sleep
== Test sixfive_test ==
$ make qemu-gdb
sixfive_test: OK (0.4s)
== Test sixfive_readme ==
$ make qemu-gdb
sixfive_readme: OK (0.6s)
== Test sixfive_all ==
$ make qemu-gdb
sixfive_all: OK (1.1s)
== Test memdump, examples ==
$ make qemu-gdb
memdump, examples: OK (1.0s)
== Test memdump, format ii, S, p ==
$ make qemu-gdb
memdump, format ii, S, p: OK (1.0s)
== Test find, in current directory ==
$ make qemu-gdb
find, in current directory: OK (1.1s)
== Test find, in sub-directory ==
$ make qemu-gdb
find, in sub-directory: OK (1.0s)
== Test find, recursive ==
$ make qemu-gdb
find, recursive: OK (1.2s)
== Test exec ==
$ make qemu-gdb
exec: OK (0.8s)
== Test exec, multiple args ==
$ make qemu-gdb
exec, multiple args: OK (1.0s)
== Test exec, recursive find ==
$ make qemu-gdb
exec, recursive find: OK (1.3s)
== Test time ==
time: OK
Score: 121/131
make: *** [grade] Error 1
(base) hamza@macbookpro xv6-labs-2025 % make qemu

```

```
mkfs/mkfs fs.img README user/findtest.sh user/sixfive.txt user/_cat user/_echo user/_forktest user/_  
_grep user/_init user/_kill user/_ln user/_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_use  
rtests user/_grind user/_wc user/_zombie user/_logstress user/_forphan user/_dorphan user/_sleep use  
r/_sixfive user/_memdump user/_find user/_memdump  
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000  
balloc: first 1164 blocks have been allocated  
balloc: write bitmap block at sector 46  
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global  
virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-devic  
e,drive=x0,bus=virtio-mmio-bus.0
```

xv6 kernel is booting

hart 2 starting

hart 1 starting

init: starting sh

\$ sleep 50

\$ sixfive sixfive.txt

5

100

18

6

\$ memdump

Example 1:

61810

2025

Example 2:

a string

Example 3:

another

Example 4:

0xcd0

cd0

1819438967

100

z

xyzzzy

Example 5:

hello

w

o

r

l

d

\$ echo > b

\$ mkdir a

\$ echo > a/b

\$ mkdir a/aa

\$ echo > a/aa/b

\$ find . b

./b

./a/b

./a/aa/b

\$ find . wc -exec echo hi

hi ./wc

\$ sh < findtest.sh

\$ mkdir: a failed to create

\$ \$ \$ \$ \$ hello

hello

hello

\$ \$